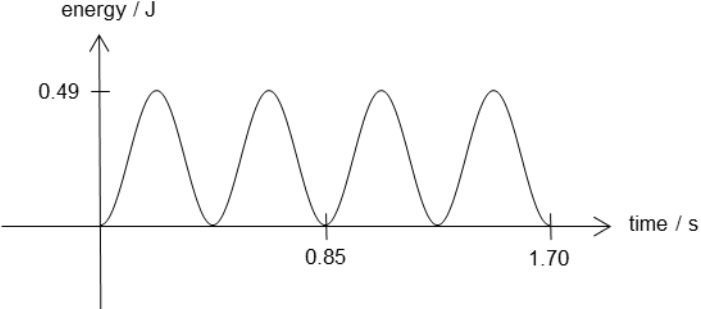
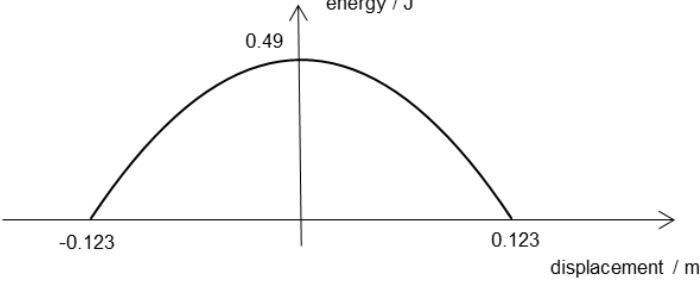
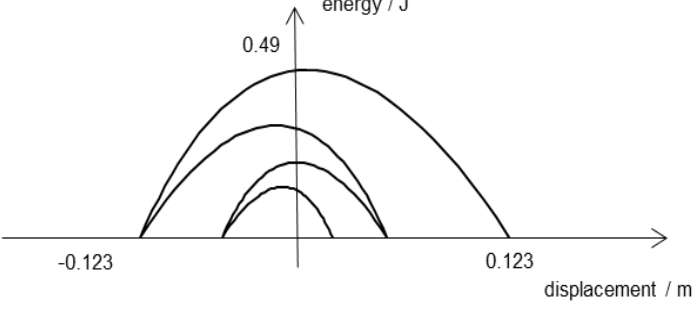


|      |                                                                                      |
|------|--------------------------------------------------------------------------------------|
| 4)a) | $F = kx$<br>$8.0 = 65x$                                                              |
|      | $x = 0.123\text{m}$                                                                  |
| b)i) | $KE_{\max} = PE_{\max}$                                                              |
|      | $KE_{\max} = PE_{\max} = \frac{1}{2}kx^2 = \frac{1}{2}(65)(0.123)^2 = 0.49\text{J}$  |
| ii)  | $KE_{\max} = \frac{1}{2}mv_o^2 = \frac{1}{2}m(\omega x_o)^2 = 0.49\text{J}$          |
|      | $\frac{1}{2}(1.2)\left(\frac{2\pi}{T}(0.123)\right)^2 = 0.49$<br>$T = 0.855\text{s}$ |
| iii) |     |
| iv)  |  |
| v)   |  |
|      |                                                                                      |
|      |                                                                                      |
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